KYLE C. M. SMITH

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EDUCATION

Aug 2025 **Ph.D., The Pennsylvania State University**
Department of Anthropology

May 2022 **M.A., The Pennsylvania State University**
Department of Anthropology

May 2023 **Postbaccalaureate Certificate, The Pennsylvania State University**
Geographic Information Systems

May 2016 **B.S. with Distinction, Duke University**
Majors: Evolutionary Anthropology, Biology

EMPLOYMENT

2025-pr. **Postdoctoral Research Scholar (P.I.: Dr. Margaret Gruen)**
Department of Clinical Sciences, North Carolina State University, Raleigh, NC

2022-25 **Instructor**
Department of Anthropology, The Pennsylvania State University, State College, PA

2022-25 **Freelance editor for scientific papers by foreign academics**
HighEdit Company, Haikou, Hainan, China

2020-22 **Teaching Assistant**
Department of Anthropology, The Pennsylvania State University, State College, PA

2017-19 **Lab Manager / Associate in Research (P.I.: Dr. Brian Hare)**
Duke Canine Cognition Center, Duke University, Durham, NC

2012-16 **Tour Guide**
Duke Lemur Center, Duke University, Durham, NC

2013-14 **Research Assistant (P.I.: Dr. Anne Pusey)**
Department of Evolutionary Anthropology, Duke University, Durham, NC

PUBLICATIONS

in press Gautam, A., Bhandari, D., Gurung, K., Gyawali, A., Gurung, K. Yadav, P., **Smith, K.C.M.**, Ahmad, A.R., Shrestha, D., Ange-van Heugten, K., Weyrich, L., Karna, A.K., Jha, A.R. Industrialization restructures the domestic dog gut microbiome while preserving host specificity. *mSystems*. Preprint on bioRxiv. https://doi.org/10.64898/2026.06.29.735155

2026 **Smith, K.C.M.** Why anthropologists should study dogs. *Evolutionary Anthropology*. https://doi.org/10.1002/evan.70050

2023 Salomons, H., **Smith, K.C.M.**, Callahan-Beckel, M., Callahan, M., Levy, K., Kennedy, B.S., Bray, E.E., Gnanadesikan, G.E., Horschler, D.J., Gruen, M., Tan, J., White, P., vonHoldt, B.M., MacLean, E.L., and B. Hare. Response to Hansen Wheat et al.: Additional analysis further supports the early emergence of cooperative communication in dogs compared to wolves raised with more human exposure. *Learning & Behavior*. https://doi.org/10.3758/s13420-023-00576-2

2021 Salomons, H., **Smith, K.C.M.**, Callahan-Beckel, M., Callahan, M., Levy, K., Kennedy, B.S., Bray, E.E., Gnanadesikan, G.E., Horschler, D.J., Gruen, M., Tan, J., White, P., vonHoldt, B.M., MacLean, E.L., and B. Hare. Cooperative communication with humans evolved to emerge early in domestic dogs. *Current Biology*. https://doi.org/10.1016/j.cub.2021.06.051

CONFERENCE PRESENTATIONS AND POSTERS

2026 **Smith, K.C.M.**, Gruen, M., and B. Hare. A comparison of dog cognition tasks conducted by citizen scientists and laboratory researchers. Presentation. Canine Science Conference, Vancouver, BC, Canada.

2026 Creamer, M., Ferrans, M., Arbeeva, L., Enomoto, M., Moore, K., Mahs, S., Woods, V., Carter, S., **Smith, K.**, Keefe, F., Hare, B. Validation of outcomes for assessing emotional and cognitive domains affected by naturally-occurring osteoarthritis pain in dogs: preliminary findings. Poster. Osteoarthritis Research Society International World Congress on Osteoarthritis, West Palm Beach, FL.

2024 **Smith, K.C.M.** A comparison of measures of fearfulness and stress in companion dogs. Presentation. Canine Science Conference, Seattle, WA.

2023 **Smith, K.C.M.** Companion dogs as a model for the impact of social relationships on stress and fitness. Presentation. 92nd Annual Meeting of the American Association of Biological Anthropologists (AABA), Reno, NV.

2022 **Smith, K.C.M.** Do cognitive tasks predict dogs' social behavior in a dog park? Presentation. Canine Science Conference, Hamilton, NY.

2021 **Smith, K.C.M.** Using camera traps to study canid behavior in Australia. Presentation. East Coast Workshop on Canine Cognition, online.

2020 **Smith, K.C.M.** Dog-human social relationships in Aboriginal Australia. Presentation. East Coast Workshop on Canine Cognition, New Haven, CT.

2018 **Smith, K.C.M.** A comparison of wolf and dog social cognition. Presentation. East Coast Workshop on Canine Cognition, New Haven, CT.

2016 **Smith, K.C.M.**, Amoroso, C., and C.L. Nunn. Water choices and disease: how do lemurs avoid exposure to parasites? Poster. Duke University Lemur Center 50th Anniversary Scientific Symposium & Gala, Durham, NC.

2016 Amoroso, C.R., **Smith, K.C.M.**, Frink, A.G., and C.L. Nunn. Parasite risk influences the water preferences of lemurs. Poster. 85th Annual Meeting of the American Association of Physical Anthropologists (AAPA), Atlanta, GA.

RESEARCH EXPERIENCE

2025-pr. **Thinking Pets Program**, North Carolina State University, Raleigh, NC
Developed tasks to measure dog cognition. Conducted power analyses to inform research design. Collected cognitive data from dogs. PI: Dr. Margaret Gruen

2021-25 **Centre County Canine Cognition Project**, Penn State University, State College, PA
Developed dog behavior project involving behavioral observations of dogs at a dog park, cognitive experiments, owner surveys, salivary and hair cortisol sampling, and dog and microbial genetic sampling. Extracted and amplified DNA for sequencing. Got approvals from IACUC, IRB, and local government. Analyzed data and presented results.

2017-19 **Duke Canine Cognition Center**, Duke University, Durham, NC
Conducted cognitive experiments with pet dogs (*Canis familiaris*). Trained and supervised team of undergraduate research assistants. Coded research videos. P.I.: Dr. Brian Hare

2017-19 **Duke Puppy Kindergarten**, Duke University, Durham, NC
Oversaw the creation, development, and management of the Duke Puppy Kindergarten program. Designed system for selecting, scheduling, and managing personnel. Trained, supervised, and managed over 120 volunteers. Oversaw research with the program's five puppies. P.I.: Dr. Brian Hare

2017-18 **Wildlife Science Center**, Stacy, MN
Designed and conducted cognitive experiments with captive wolves (*Canis lupus*). Prepared blood samples for hormone and DNA analysis. P.I.: Dr. Brian Hare

2016 **Project Howler Monkey Palenque**, Palenque, Mexico
Worked with multinational team to track and study spatial navigation of wild black howler monkeys (*Alouatta pigra*) using scan sampling. Marked feeding trees. P.I.: Dr. Sarie Van Belle and Dr. Alejandro Estrada

2014-16 **Duke Lemur Center**, Duke University, Durham, NC
Conducted independent research on lemurs' disease avoidance behaviors. Designed study, collected and analyzed data, and wrote and defended honors thesis. P.I.: Dr. Charles Nunn. Also mentored by Ph.D. student (now Dr.) Caroline Amoroso

2014 **Organization for Tropical Studies**, Kruger National Park, South Africa
Conducted field research and wrote papers on various topics, including invasive mollusks, and human-wildlife conflict, as part of semester courses on savanna ecology and conservation. Learned field research methods, including mist netting, vegetation transects, camera trapping, biodiversity surveys.

2014 **Duke Lemur Center SAVA Conservation Initiative**, Sava Region, Madagascar
Tracked and studied bamboo lemurs (*Hapalemur occidentalis*) using group scan sampling. Flagged and GPS mapped trails. Radio-tracked dwarf lemurs (*Cheirogaleus*). P.I.: Dr. Erik Patel

2013-14 **Jane Goodall Institute Research Center**, Duke University, Durham, NC
Extracted and digitized data from chimpanzee (*Pan troglodytes*) focal follow sheets that were collected in Gombe National Park, Tanzania. P.I.: Dr. Anne Pusey

GRANTS AND FELLOWSHIPS

2024	Center for Human Evolution and Diversity Seed Grant, Penn State	$5064
2023	Research and Graduate Studies Office Dissertation Support, Penn State	$1980
2023	Pollitzer Student Travel Grant, AABA	$500
2023	Center for Human Evolution and Diversity Seed Grant, Penn State	$5026
2022	RGSO Conference Travel Award, Penn State University	$400
2022	Hill Fellowship Pre-Comprehensive Award, Penn State University	$2000
2021	Tygel Endowment for Educational Programs, Penn State University	$516
2021	Honorable Mention for NSF Graduate Research Fellowship Program	N/A
2021	Flower Grant for Ecological Research, Penn State University	$4000
2020	External Funding Incentive Award, Penn State University	$500
2020	Hill Fellowship Training Award, Penn State University	$1220
2019-20	University Graduate Fellowship, Penn State University	$30000

2015	Deans' Summer Research Fellowship, Duke University	$3000
2015	Undergraduate Research Support Independent Study, Duke University	$400
2015	Molly Glander Research Award, Duke Lemur Center	$300

TEACHING EXPERIENCE

2023-25 Instructor, Pennsylvania State University, World Campus
ANTH 21: Introduction to Biological Anthropology (five semesters)

2022-23 Instructor, Pennsylvania State University, World Campus
ANTH 45N: Cultural Diversity (three semesters)

2020-23 Teaching Assistant/Guest Lecturer, Pennsylvania State University, State College, PA
ANTH 21: Introduction to Biological Anthropology (three semesters)

2022 Teaching Assistant/Guest Lecturer, Pennsylvania State University, State College, PA
ANTH 1: Introduction to Anthropology (one semester)

2021 Teaching Assistant, Pennsylvania State University, State College, PA
ANTH 140: Anthropology of Alcohol (one semester)

2013 Student Instructor, Duke University, Durham, NC
HOUSECS 59: Created Worlds (one semester)

MENTORING EXPERIENCE

2022-24 Mentored undergraduate students involved in research in dogs, including a senior honors thesis student and students who presented poster at undergraduate symposium. Held weekly meetings for research discussions and professional development. Penn State University, State College, PA

2020-21 Mentored anthropology undergraduate through the student-led Graduate-Undergraduate Mentorship Program. Penn State University, State College, PA

2019 Mentored undergraduate independent study students. Duke Canine Cognition Center, Durham, NC

2013 ESL Tutor, George Watts Montessori Magnet School, Durham, NC

2013 ESL Tutor, Durham School of the Arts, Durham, NC

OUTREACH

2026 **North Carolina Museum of Natural Sciences**, Raleigh, NC
Presented about dog cognition research at multiple science outreach events, performing live demonstrations with dogs.

2021-24 **Stories Under the Sky**, State College, PA
Co-developed and presented at recurring public talk series (over 20 events, with hundreds of attendees total) on the anthropology of stargazing, giving lectures on various topics and arranging guest storytellers. Led programming in both English and Spanish.

2023-24 **AnthroFest**, Penn State, State College, PA
Co-developed anthropology-themed outreach event, which had over 300 attendees. Developed and led one of the activities at the event (about dog domestication). Managed volunteers.

2023 **Anthropological Research in Science Education**, Penn State, State College, PA
Helped organize two-week intensive summer research program for underrepresented undergraduate students, funded by the NSF. Mentored students in independent projects using GIS.

2023 **Anthropology field trip in Spanish for high schoolers**, Penn State, State College, PA
Organized and co-led Spanish language fieldtrip of an English Language Learner (ELL) high school class to visit Penn State's anthropology department and museum.

2022 **STEM Night**, Radio Park Elementary School, State College, PA
Gave interactive live demonstrations of dog cognition research. Ran educational game about domesticated animals. Led two undergraduate volunteers.

2022 **Microbe activity in Spanish**, State College High School, State College, PA
In a team, guided an English Language Learner (ELL) class in Spanish in a hands-on activity of data collection and experimental design involving microbe culturing.

2022 **Ask a Scientist**, Grange Fair, Centre Hall, PA
Gave interactive live demonstrations of dog cognition research. Created educational game about domesticated animals.

2022 **Talk and demo on dog cognition**, JCC Nursery School of State College, PA
Talked about dogs, domestication, and science with 3 to 5 year old summer camp attendees. Did live demonstrations with a dog to show how to study how dogs think.

2022 **Science Pub**, Big Spring Spirits, Bellefonte, PA
Gave talk "How dogs became our best friends" to members of the public.

2021-22 **Cherry Springs State Park**, PA
Set up telescopes and guided visitors in viewing astronomical objects.

2020-23 **Skype a Scientist**
Presented remotely about dogs, primates, and anthropology to elementary school students in the US and abroad.

2017-19 **Duke Canine Cognition Center**, Duke University, Durham, NC
Presented and demonstrated canine cognition research to visitors to lab.

2012-16 **Duke Lemur Center**, Duke University, Durham, NC
Worked as tour guide to provide scientific and environmental education about lemurs to public, including children and adults.

PROFESSIONAL SERVICE

2023-24 President, Anthropology Graduate Student Association, Penn State

REVIEWER

Journal of Zoology
Biology Letters

SKILLS

Languages:
- English (native)
- Spanish (intermediate)
- French (basic)

Software:
- R statistical software
- ArcGIS Pro
- Microsoft Word, Excel, and PowerPoint
- Behavioral Observation Research Interactive Software (BORIS)
- QIIME2

Wilderness and backcountry skills and first aid